

Active Learning for Finding a Needle in a Haystack

Anonymous ACL submission

Abstract

TBD

1 Introduction

A really nice motivating example throughout this is hate speech

- Very small amount of speech is hate speech: <https://phys.org/news/2023-04-analysis-speech-significantly-twitter.html> - so it's a needle in a haystack problem
- There are lots of different types of hate speech - some of them are big and some of them are small
- But we're interested in finding all of them, even if they make up a very small part of total hate speech
- So we're interested in this "bias" (defined below) idea rather than just accuracy in finding hate speech overall

More generally, needle in a haystack problems must be common, because generally the most applicable cases for AL are those with a small target class - if it's easy to find positives, then AL is less important.

[We've reformulated the problem a bit since I wrote this] There are two problems that we're trying to solve with Active Learning

1. Biased seed - don't want to this to affect our final classification
2. Imbalanced data - want to disproportionately label the target class (ie. want labelling to be efficient)

Random/DAL does really well on the former, but badly on the latter

- DAL experiments with highly imbalanced data, we find that it pulls out close to 0 of the target class for labelling
- Somewhat unsurprising as in both cases, you are making your training sample representative of your pool and your pool is really unbalanced, so it's going to be really inefficient

Uncertainty-based methods do well on the latter, but poorly on the former - they are known for introducing bias <https://arxiv.org/pdf/1909.09389.pdf>

So if we set this as the goalposts, we can't have efficient, representative active learning

A goal that makes more sense: we want the positive labels to be representative of all the positives in the data and the negatives to be representative of all the negatives in the data, while having a roughly equal number of positives and negatives.

2 Literature

Previous active learning lit <https://arxiv.org/abs/2007.00077> - focuses on rare classes

Surveys, Active Learning with BERT

Also related to the papers on domain adaptation

- Melissa/Jake suggested this: <https://arxiv.org/pdf/2403.03208.pdf>
- <https://melissadell.atlassian.net/wiki/spaces/TCC/pages/2584412161/Domain+Adaptation+Discriminative+Active+Learning>
- <https://spj.science.org/doi/10.34133/icomputing.0058#abstract>
- https://link-springer-com.ezp-prod1.hul.harvard.edu/chapter/10.1007/978-3-031-33377-4_39
- <https://aclanthology.org/D18-1318/>

069	• https://arxiv.org/abs/2203.13450	3.2.1 Looking at mispredictions	111
070	• https://arxiv.org/abs/2210.10109	An ideal measure: run the AL method thousands of times, with different seeds. Look at whether there are some data points that are more likely to be mispredicted than others. But this is not compute feasible!	112
071	• https://proceedings.neurips.cc/paper_files/paper/2022/file/8c64bc3f7796d31caa7c3e6b969bf7da-Paper-Conference.pdf	Another measure (less ideal, more feasible): somehow chunk the space up and see whether there are systematically different accuracies/F1s in different parts of the space.	113
072			114
073			115
074			116
075	• https://dl.acm.org/doi/abs/10.1145/3510414		117
076			118
077	• https://www-sciencedirect-com.ezp-prod1.hul.harvard.edu/science/article/abs/pii/S0950705122009248	3.2.2 Exploring the space	119
078		If all points are a priori equally difficult to classify, then we want a uniform sampling of the support of the distribution of the pool. (If they're not equally difficult to classify, then we'd want this to be proportional to their difficulty of classification).	120
079		Related to the idea of whether we have explored the space.	121
080	• https://aclanthology.org/2020.emnlp-main.637/	Ways of measuring this:	122
081		• Time taken (number of iterations) to find rest of the positive class	123
082	• https://aclanthology.org/2021.emnlp-main.51.pdf	• Diversity	124
083		– Every point as a labelled point that is close to it	125
084	• https://aclanthology.org/2021.findings-emnlp.36v2.pdf	– AL for BERT does this by using Euclidean distance on the CLS token	126
085		– They take the mean of the minimum distance	127
086	3 Measures/metrics/whatever	– Might want to think about taking the max of the minimum distance - could be a better measure of whether some spaces are entirely missed	128
087	3.1 Measuring efficiency	• Are there other ways of looking whether a sample has labels for the support of the distribution?	129
088	• Number of on-topic and off-topic articles pulled out	– Something to do with evenly spaced labels?	130
089		– Average distance between labelled point?	131
090	• Accuracy, F1, Precision, Recall (AL with BERT paper comments on improvements on precision and recall)	• Given the goal outlined above, it seems to make sense so look at these each class at a time, or just for the positive class, rather than across the whole dataset.	132
091			133
092			134
093	• Number of iterations until all positives are labelled		135
094			136
095	• Representativeness - For each labelled sample, how close are the 10 nearest unlabelled point - measures whether you're disproportionately picking outliers. This isn't really a bias-related measure, but an efficiency-related measure. Not very sure this is important, because we have better measures of efficiency.		137
096			138
097			139
098			140
099			141
100			142
101			143
102	3.2 Measuring bias		144
103	What do we mean by bias? Unbiased = errors made by the model are as good as random. Ie. every article in the dataset has the same probability of being mispredicted.		145
104	Using the hate speech example, there is no marginalised group where we are systematically more likely to mispredict hate speech.		146
105	How to measure this?	More/older thoughts on bias, and things that people have previously written about this:	147
106		We could have:	148
107			149
108			150
109			151
110			152

1. Bias from the initial seed - we take this as given
2. Bias from the AL method - ideally we want this to eliminate the bias from the seed and not introduce any additional bias, but there's evidence that some AL methods increase bias
 - <https://arxiv.org/pdf/1909.09389.pdf> divide this kind of bias into two types:
 1. Class bias: Greedy uncertainty based query strategies are said to pick disproportionately from a subset of classes per query (Sener and Savarese, 2018; Ebert et al., 2012), developing a lopsided representation in each query. However, its effect on the resulting sample set is not clear. We test this by measuring the Kullback-Leibler (KL) divergence between the ground-truth label distribution and the distribution obtained per query as one experiment and over the resulting sample ($\cap S$) as the second.
 - Note: this is really not what we want - we want balance between classes!
 - So we actually want to exact opposite of this measure
 2. Feature bias
 - Uncertainty sampling can lead to undesirable sampling bias in feature space (Settles, 2009) by repeating redundant samples and picking outliers (Zhu et al., 2008). Diversity-based query strategies (Sener and Savarese, 2018) are used to address this issue, by selecting a representative subset of the data.
 - They find that it is biased, but it is biased in a way that helps with classification because it picks things close to the boundary
 - Note: but this is bad if the boundary is wrong - if we have a biased seed, the initial boundary could be quite wrong, and if we only ever test things close to the boundary, we will never get over this bias

4 Our method

Basic idea: split pool into predicted positives and predicted negatives and then run DAL

Two versions of this:

1. Run DAL among pool of predicted positives, pulling $n/2$ samples, then repeat among pool of predicted negatives
2. Reweight the pool so the number of predicted positives and predicted negatives are the same, then run DAL in this pool

In practice, the two give similar results.

4.1 Stopping Criterion AL

NIHDAL lends itself to a stopping procedure - when you are getting consistently 50% on topic and not, then you are done ... I think ?

Alternative stopping criterion: when you get down to nothing returned as positive

5 Datasets

We evaluate on three publicly available text classification benchmarks, all sourced from the Hugging Face Hub. They are deliberately different in topic and class structure.

AG News (Zhang et al., 2015) is a four-way news topic classification dataset whose classes are World, Sports, Business and Sci/Tech. The Hugging Face release ships balanced train (120,000) and test (7,600) splits, with $\sim 30,000$ and 1,900 examples per class respectively.

TREC-10 (Li and Roth, 2002; Hovy et al., 2001) is a six-way coarse-grained question-type classification dataset whose classes are ABBR (abbreviations), ENTY (entities), DESC (descriptions), HUM (humans), LOC (locations) and NUM (numerics). The Hugging Face release ships 5,452 training questions and 500 test questions; class sizes are imbalanced, with ABBR the smallest at ~ 86 training examples.

UCB Measuring Hate Speech (Sachdeva et al., 2022) is a corpus of social-media comments annotated by multiple annotators along several dimensions, including an ordinal hate-speech-intensity score (hate_speech_score) and indicator flags for which demographic groups each comment targets (race, religion, gender, sexuality, age, disability, origin) and which sub-groups within them (e.g. Muslim, Jewish, Christian under religion). The Hugging Face release contains 135,556 annotator-level rows over 39,565 unique comments and ships only a train split.

6 Experiments

6.1 Active Learning Strategies tested

Things that we experiment with:

- Random
- Uncertainty-based
 - Least confidence (Lewis and Gale, 1994)
 - BALD (Houlsby et al., 2011; Gal and Ghahramani, 2016)
- Gradient-based, aka, expected model change
 - EGL (Huang et al., 2016) - not doing this, small-text guy said no one uses it any more
 - BADGE (Ash et al., 2019)
- Distribution-based aka diversity sampling
 - Core set (Sener and Savarese, 2018; Geifman and El-Yaniv, 2017)
 - DAL (Gissin and Shalev-Shwartz, 2019)
 - Contrastive active learning (?) - have not currently run this. Should look into it more
- Hybrid/Ensemble results
 - Haven’t done any of these. Should read up more about these

Ours is either distribution-based or hybrid.

We might want to think about whether we can apply our split-sample approach to other methods. We can predict positives and negatives and then sample randomly within each. I guess we could do this with least confidence as well?

6.2 Data preparation

Across all three benchmarks (Section 5) we apply a common data-preparation pipeline.

(a) Binarisation. For each dataset we choose a target class and collapse the multi-class label to $\{0, 1\}$, with 1 marking the target class and 0 everything else. The target class for each dataset is given in Table 1.

(b) Down-sampling to $\sim 1\%$ positive rate. If the natural positive rate is materially above 1% we down-sample the positive class until it accounts for 1% of the dataset; if it is already at or near 1%, no down-sampling is applied. The same rate is applied

to train and test. This puts the target class firmly into “needle in a haystack” territory, which is the regime where active learning is most useful.

(c) Biased and unbiased configurations. For each dataset we run experiments under *two* warm-start configurations. The biased configuration is motivated by a common practical pattern: a labeller starting a project is often aware of only one part of the target class, and we want to know whether an AL method can recover the parts they have not yet seen. The unbiased configuration is a control.

Both configurations produce an initial labelled set of 100 examples (50 positive, 50 negative); negatives are always sampled uniformly at random, and the configurations differ only in how positives are drawn. In the *unbiased* configuration the 50 positives are sampled uniformly at random from the full train positive pool. In the *biased* configuration they are sampled uniformly at random from a designated *seed sub-population* of the positives; all other positives are excluded from initialisation and tracked through the AL loop as a recovery diagnostic. The target class, seed sub-population, and resulting dataset sizes for each dataset and configuration are given in Table 1.

For UCB Measuring Hate Speech the held-out sub-population is naturally available *within* the target class — the corpus annotates which religious sub-group each comment targets — so we can hold out non-Muslim religion-targeted positives without changing what counts as a positive at all. AG News and TREC-10 do not have natural sub-populations within a single class, so to construct a comparable biased setup we widen the target class to a union of two neighbouring classes ($\text{World} \cup \text{Sports}$, $\text{ABBR} \cup \text{ENTY}$) and treat the second class as the held-out sub-population. This is why the biased target class is strictly broader than the unbiased one for those two datasets, and why their post-preprocessing dataset sizes differ between configurations in Table 1, while the two hate-speech rows have identical sizes by construction.

7 Early Stopping in training

Unlike the original DAL paper we do not find that early stopping is helpful in the discriminative training.

8 Limitations

Should extend beyond binary classification? But we haven’t implemented

Dataset	Setup	Target class	Positive seed pool	Train size	Test size	% target (train)
AG News	Unbiased	World	World	90,909	5,757	1.00%
	Biased	World $\cup$ Sports	World	60,606	3,838	1.00%
TREC-10	Unbiased	ABBR	ABBR	5,420	495	1.00%
	Biased	ABBR $\cup$ ENTY	ABBR	4,157	401	0.99%
UCB Hate Speech	Unbiased	Religion targeted	Religion-targeted	31,587	7,901	1.34%
	Biased	Religion targeted	Muslim-targeted	31,587	7,901	1.34%

Table 1: Benchmark datasets after data preparation (Section 6.2), shown for both experimental configurations. “Positive seed pool” is the subset of train positives from which the warm-start sample draws its 50 positive examples; for the biased configuration of AG News and TREC-10 the positive class is widened so that the held-out sub-population (Sports / ENTY) is part of the AL pool but not the seed. Hate speech keeps the same target class across configurations, with the seed pool restricted to Muslim-targeted comments in the biased configuration. The target down-sampling fraction is 0.01 for AG News and TREC-10; hate speech is left at its natural rate ($\sim 1.34\%$) because annotator aggregation already yields close to that target. Hugging Face source identifiers and full per-dataset processing details are given in Appendix B.

Acknowledgments

References

- Jordan T. Ash, Chicheng Zhang, Akshay Krishnamurthy, John Langford, and Alekh Agarwal. 2019. [Deep batch active learning by diverse, uncertain gradient lower bounds](#). *CoRR*, abs/1906.03671.
- Yarin Gal and Zoubin Ghahramani. 2016. [Dropout as a bayesian approximation: Representing model uncertainty in deep learning](#). In *Proceedings of The 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–1059, New York, New York, USA. PMLR.
- Yonatan Geifman and Ran El-Yaniv. 2017. [Deep active learning over the long tail](#). *CoRR*, abs/1711.00941.
- Daniel Gissin and Shai Shalev-Shwartz. 2019. [Discriminative active learning](#). *CoRR*, abs/1907.06347.
- Neil Houlsby, Ferenc Huszár, Zoubin Ghahramani, and Máté Lengyel. 2011. [Bayesian active learning for classification and preference learning](#). *Preprint*, arXiv:1112.5745.
- Eduard Hovy, Laurie Gerber, Ulf Hermjakob, Chin-Yew Lin, and Deepak Ravichandran. 2001. [Toward semantics-based answer pinpointing](#). In *Proceedings of the First International Conference on Human Language Technology Research*.
- Jiaji Huang, Rewon Child, Vinay Rao, Hairong Liu, Sanjeev Satheesh, and Adam Coates. 2016. [Active learning for speech recognition: the power of gradients](#). *CoRR*, abs/1612.03226.
- David D. Lewis and William A. Gale. 1994. [A sequential algorithm for training text classifiers](#). *CoRR*, abs/cmp-lg/9407020.
- Xin Li and Dan Roth. 2002. [Learning question classifiers](#). In *COLING 2002: The 19th International Conference on Computational Linguistics*.

- Pratik S. Sachdeva, Renata Barreto, Geoff Bacon, Alexander Sahn, Claudia von Vacano, and Chris Kennedy. 2022. The measuring hate speech corpus: Leveraging rasch measurement theory for data perspectivism. In *Proceedings of the 1st Workshop on Perspectivist Approaches to NLP @LREC2022*.
- Ozan Sener and Silvio Savarese. 2018. [Active learning for convolutional neural networks: A core-set approach](#). *Preprint*, arXiv:1708.00489.
- Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, volume 28.

A Implementation notes

Unlike the original DAL paper we do not find that early stopping is helpful in the discriminative training.

B Dataset-specific preprocessing

The shared preprocessing pipeline (binarise $\rightarrow$ down-sample to $\sim 1\%$) is described in the main body. Per-dataset choices are below.

AG News. Used as-is from the Hugging Face `ag_news` release: 120,000 train and 7,600 test examples, balanced 25% per class. The target class is selected by the loader’s `target_labels` argument; the default `(0,)` corresponds to *World*. For biased-seed experiments, `target_labels = [0, 1]` so that the second listed class (*Sports*) acts as the held-out sub-population: those rows are excluded from the initial seed and tracked through the AL loop.

TREC-10. Loaded from the Hugging Face `trec` release, renaming the `coarse_label` column to `label`. There are 5,452 train and 500 test questions,

with the canonical six-way coarse-grained label set (ABBR, ENTY, DESC, HUM, LOC, NUM); ABBR is the smallest coarse class with a natural rate of $\sim 1.6\%$ in train. The default target class is ABBR (`label = 0`); for biased-seed experiments, `target_labels = [0, 1]` so that ENTY is held out.

UCB Measuring Hate Speech. Loaded from the Hugging Face `ucberkeley-dlab/measuring-hate-speech` release. Requires substantially more preprocessing than the other two benchmarks:

1. **Annotator aggregation.** The raw release contains one row per (comment, annotator) pair. We group by (`comment_id`, `text`) and average the numeric fields (overall `hate_speech_score`, `target_religion`, and each of the eight per-religion indicators), reducing 135,556 rows to 39,565 unique comments.
2. **Label derivation.** The binary label is set to 1 where the aggregated `hate_speech_score > 1` and aggregated `target_religion = 1`; 0 otherwise.
3. **Per-religion strict flags.** For each of the eight religious groups (atheist, buddhist, christian, hindu, jewish, mormon, muslim, other) we form a strict flag equal to 1 where `target_religion_X = 1` and `hate_speech_score > 1`.
4. **Inconsistency filter.** 77 rows where the binary label is 1 but no strict religious flag is set are dropped, leaving 39,488 comments.
5. **Stratification column.** Each row is assigned a `strat_col` equal to the *smallest* religious sub-population it strictly targets (resolving ties by religion-size order); rows that target no religion are labelled “none”. This rare-first assignment ensures that the smallest sub-populations are represented in both splits.
6. **Train/test split.** The Hugging Face release ships only a train split, so we form an 80/20 split stratified on `strat_col` with fixed `random_state = 42`.
7. **No down-sampling.** Because annotator aggregation already yields a positive rate close to the 1% target, we apply no further imbalance.

The `target_fraction` argument is accepted by the loader for registry compatibility but ignored.

8. **Biased sub-population.** For biased-seed experiments, the seed pool is restricted to *Muslim-targeted* positives (the largest religious sub-population, $n = 184$ in train); the other 240 religion-targeted train positives are returned as `bias_indices` and held out from initialisation.

C Motivating econometrics/statistics

Problem Statement. Suppose we have a binary variable $D \in \{0, 1\}$ and a high-dimensional covariate vector $\mathbf{X} \in \mathcal{X} \subseteq \mathbb{R}^d$. The *true* D is partially or fully unobserved, and instead we only have access to a *predicted or measured* variable

$$\hat{D} = \hat{D}(\mathbf{X}),$$

which may be subject to misclassification relative to D . We define the misclassification indicator

$$M = \mathbf{1}\{\hat{D} \neq D\}.$$

This setup arises in numerous applications:

- **Classification tasks in machine learning (ML) or computer science (CS):** $\hat{D}$ is learned from data, potentially with high-dimensional features (e.g. text embeddings, images).
- **Measurement error in econometrics or statistics:** D is a latent binary variable (e.g. “treatment” or “eligibility”), and $\hat{D}$ is a noisy observed proxy.

Motivation: Downstream Tasks and Shortcomings of Overall Accuracy. Standard practice in computer science and ML is to evaluate a classifier $\hat{D}(\mathbf{X})$ by *overall accuracy*, precision, recall, AUC, or other aggregate predictive metrics. However, these metrics *do not* fully capture whether misclassification is *differential with respect to $\mathbf{X}$* , i.e. if there exist subpopulations (possibly small) in which misclassification rates deviate systematically from the global average.

This distinction matters especially when there are *downstream tasks* such as regression or causal inference, where D plays an important role (eg as a selection variable or a regressor). If $\hat{D}$ makes *systematic* errors in subpopulations of $\mathbf{X}$ —in other

words, $\Pr(\hat{D} \neq D \mid \mathbf{X} = x)$ depends on x —it can result in additional bias that could be significantly than for instance standard "attenuation" bias from random measurement error.

Basic example 1 If D selects instances for treatment effect estimation and it systematically selects the ones with low treatment effects that would yield a significant bias in treatment effect estimation.

Could write a more explicit example based on regression like this A canonical downstream task is the linear model

$$Y = \alpha + \beta D + \gamma^\top \mathbf{X} + u,$$

with $\mathbb{E}[u \mid D, \mathbf{X}] = 0$. If we replace D by $\hat{D}$ in a naive regression of Y on $(\hat{D}, \mathbf{X})$, bias arises whenever $\hat{D}$ is systematically correlated with u . Differential misclassification with respect to $\mathbf{X}$ —especially in regions where u or β is large—can induce significant bias. In contrast, if $\hat{D} \neq D$ is *nondifferential* w.r.t. $\mathbf{X}$, the main effect is an overall attenuation of β rather than a more severe bias.

Measuring Differential Misclassification

So clearly we want some measures of differential misclassification.

The first basic one which we should definitely have because it is nice and intuitive is simply the F1 or any other classic CS score on a small subclass of the target class (for instance the unseeded one).

But then we can think about some more formal options.

Testing the Constancy of Conditional Misclassification Probability

A natural approach is to examine

$$m(\mathbf{x}) = \Pr(M = 1 \mid \mathbf{X} = \mathbf{x})$$

If $m(\mathbf{x})$ is *constant* in $\mathbf{x}$, we say misclassification is *nondifferential* (no subpopulation has systematically higher errors). If $m(\mathbf{x})$ is not constant, we have differential misclassification.

In practice, we want a summary measure of this and we need a way of knowing whether the deviations of an estimated $m(\mathbf{x})$ are significant.

Cross-Validated Prediction & Permutation Test. In high-dimensional settings, a practical and conceptually straightforward method is:

1. **Treat M as the target to predict from $\mathbf{X}$.** Train a flexible classifier (e.g., random forests,

gradient boosting) to predict M using $\mathbf{X}$, using cross-validation for out-of-sample metrics.

2. **Compare observed predictive performance to a null/permutation distribution.** Under the null hypothesis of no dependence between M and $\mathbf{X}$, permuting the M_i labels relative to $\mathbf{X}_i$ should not decrease predictive power. Formally, compute a p-value by how often random permutations yield an equal or better out-of-sample AUC (or accuracy) than the original.

A significant result implies that $\hat{D} \neq D$ is predictable from $\mathbf{X}$, so misclassification must be differential. If it is not significant, we fail to reject nondifferential errors. Then maybe we could look at the p-value as a summary value to compare the different methods.

Why a Permutation Test Is Necessary. Merely observing that this prediction achieves an AUC above 0.5 or an accuracy above chance does not tell us *how statistically significant* the improvement is nor whether it could arise from overfitting. A *permutation test* provides a formally justified p-value in a distribution-free manner:

1. **Null hypothesis.** Under H_0 : M is independent of $\mathbf{X}$ (i.e. no differential misclassification). In such a case, random permutations of M relative to $\mathbf{X}$ should produce similar cross-validated accuracy/AUC as the original pairing.
2. **Permutation procedure.** Shuffle $(M_1, \dots, M_n)$ among the $(\mathbf{X}_1, \dots, \mathbf{X}_n)$, retrain and re-evaluate the classifier to obtain a distribution of predictive metrics under H_0 .
3. **p-value.** If the observed predictive performance (e.g. out-of-sample AUC) on the actual data is in the extreme tail of the permuted distribution, we reject H_0 at the chosen significance level, concluding $\Pr(M = 1 \mid \mathbf{X})$ is *not* constant.

Key benefit: The permutation approach is *not* just an aggregate measure of predictive performance—it gives a formal test of “is there truly a dependence between $\mathbf{X}$ and M in the population?” Even a modest AUC could be highly significant if the dataset is large, whereas a large AUC might fail to be significant if sample size is small or if M is noisy in ways

that the classifier can exploit by chance. Hence, a permutation test rigorously distinguishes a genuine systematic pattern from one that could be explained by random fluctuations or finite-sample overfitting.

Some other ideas: Shape-Testing and Treatment Effect Heterogeneity. The econometrics literature on testing shape constraints seems very relevant. Maybe we can adapt some or find some non-parametric specification tests to check if $m(\mathbf{x})$ is constant. Maybe the Max Farrell binning paper and testing shape constraints would be relevant, though I am not sure it applies in the high-dimensional setting.

There is also a pretty strong seeming relevance of the treatment effect heterogeneity literature. A basic idea would be regress D on $\hat{D}$ and see whether X makes a difference and that is testing for treatment effect heterogeneity and there has been tons of results in the past years, in particular for RCTs with high-dimensional covariates, which is more or less exactly our situation.

Considering Small Regions or Rare Subpopulations

This still doesn't address all the potential worries. A potential limitation of global tests (or average measures) is that they weigh each $(\mathbf{x}_i, M_i)$ according to the empirical distribution of $\mathbf{X}$. Hence, large errors in a *small* region of $\mathbf{X}$ may be diluted in an overall average. However, that small region could be highly relevant in downstream analysis.

Some ideas.

- **Subgroup Analysis or Bin/Cluster Approaches.** Partition $\mathbf{X}$ into subgroups (possibly via clustering). Evaluate misclassification *within* each subgroup. This identifies “hotspots” where $m(\mathbf{x})$ is large, even if the group is small.

- **Nonparametric “Anomaly” or “Hotspot” Detection.** Scan statistics or local detection methods can systematically search for small regions of high misclassification. In high dimensions, approximate methods (e.g. cluster-based scanning) are often used.

- **Sup Tests.** Really what we want is something like a uniform test, which suggests looking at the sup of the deviation.

Note that actually maybe the sup is too strong of a condition because in a region where this

basically no points it actually wouldn't matter, so maybe it would be sup with a minimal mass constraint.

Statistically, this does not seem obvious to get actual results on.

- **Treatment Effect Heterogeneity (Subgroup) Analyses.** I think there is some literature in the treatment effect literature on finding subgroups that were particularly affected (or negatively affected), which again is more or less exactly our problem.

All these strategies aim to ensure that rare but systematically misclassified subpopulations are detected.